

Multimodal Fashion & Context Retrieval

Glance ML Internship Assignment · Codebase: <https://github.com/prakhar-raj-031104/FASHIONATTIRE>

A natural-language image search engine over **3,200 fashion images** that reasons jointly about **what** a person wears, **where** they are, and the outfit's **vibe**.

Core thesis: a single global CLIP embedding is *structurally* incapable of compositional retrieval, so the fix must add **structure**, not a bigger encoder.

Corpus	Latency	Key measured result	Training
3,200 images · 9,851 garment regions · FAISS ×3 + SQLite	~150 ms warm	Colour-swap overlap@10: vanilla CLIP 0.750 → this system 0.525 (–30%, wins 8/12 pairs)	None — fully zero-shot

1. Why this is hard

#	Query	Capability	CLIP's problem
1	bright yellow raincoat	colour + garment	sparse fine-grained fashion vocabulary
2	business attire in a modern office	style + scene	needs scene grounding
3	blue shirt on a park bench	colour + garment + scene	multi-constraint conjunction
4	casual weekend outfit for a city walk	style inference	no garment words at all
5	a red tie and a white shirt	compositional binding	canonical CLIP failure

The actual failure mechanism. CLIP is trained contrastively over a **single pooled embedding per image**. Pooling is order- and position-agnostic: the vector encodes *"there is red, white, a tie, a shirt"* but **not which attribute binds to which object**. So *"red tie + white shirt"* and *"white tie + red shirt"* map to nearly the same point — literally the same bag of concepts. This is **architectural, not capacity**: scaling or fine-tuning the encoder cannot recover information destroyed at pooling. **Therefore the fix must reintroduce structure below the image level.**

2. Approaches and trade-offs (*Deliverable 1*)

Approach	Strengths	Weaknesses	Best when
A. Vanilla CLIP + FAISS	trivial, strong zero-shot	bag-of-words : no binding; weak fashion terms	quick baseline, single-concept queries
B. FashionCLIP + FAISS	large domain gain on garment/colour/fabric for ~zero cost	same architectural binding limit	any fashion task — the <i>floor</i> , not the solution
C. Caption → LLM attributes	interpretable; good scene/style	errors compound (hallucinated caption → wrong attribute); non-deterministic; extra GPU	open-vocabulary attributes with reliable captions
D. CLIP zero-shot tagging	grounded in <i>pixels</i> ; deterministic; cheap; editable vocabulary	image-level only → still no binding	structured, filterable attributes without training
E. Region decomposition	genuinely solves binding ; dataset-agnostic	segmentation cost; fails on tiny/occluded items	whenever compositionality is graded
F. Detector + scene graph + VQA, or Pinecone/microservices	maximal expressiveness	large complexity, more failure surface, no measured gain at this scale	large production systems

Decision: **B + D + E**, plus a caption signal from C (BLIP only, **no LLM extraction** — grounding attributes in pixels beats grounding them in possibly-hallucinated text). **F rejected** as over-engineering: the brief says favour ML logic

over indexing engineering, so I used FAISS + SQLite and spent the effort on ranking.

3. Chosen architecture (*Deliverable 2*)

Part A — Indexer (`src/indexing/`). Per image: (1) **FashionCLIP** global embedding → `global.faiss`; (2) **BLIP caption** → **MPNet** sentence embedding → `caption.faiss`; (3) **CLIP zero-shot attributes** (environment/style/garment/colour, each value embedded through several **prompt templates and averaged**) → **SQLite**; (4) **SegFormer garment regions**, each cropped, embedded, and tagged with **its own colour and type** → `region.faiss` + **SQLite**. Artefact (4) is the crux: it preserves *which colour sits on which garment*.

Part B — Retriever (`src/retrieval/`). *Stage 1 (recall)*: query embedded by FashionCLIP + MPNet; union of top-N from the global and caption indexes forms a 400-candidate pool. *Stage 2 (rerank)*: only that pool is scored with expensive signals. This ANN → rerank split is why the logic is unchanged at 1M images (§6).

```
score = 0.40·global_clip + 0.20·caption_sim + 0.20·attribute_match + 0.20·region_binding
```

Signal	Fixes	Rescues
<code>global_clip</code>	overall semantics + scene	all
<code>caption_sim</code>	style inference — caption "a man in a hoodie walking down a street" matches "casual weekend outfit for a city walk" with zero shared garment words	4
<code>attribute_match</code>	grounded multi-attribute filtering	1, 2
<code>region_binding</code>	compositional colour ↔ garment binding	5

Query parsing. A deterministic rule + synonym + adjacency parser yields a `QuerySpec`: structured attributes plus (colour, garment) **bindings** ("a red tie and a white shirt" → [(red,tie), (white,shirt)]). Rules over an LLM because the schema is small and fixed, it is instant, reproducible and unit-testable — while remaining a drop-in point for an LLM parser (§7.2).

Binding score. A binding scores high only if **one single region** matches *both* garment type *and* colour. Multiple bindings combine as `0.5·mean + 0.5·min`: the query is a **conjunction**, so half-satisfaction must not score like full (plain `mean` is too lenient), but plain `min` is brittle — one missed segmentation would zero a perfect image.

Two subtleties, both found by measurement: 1. **Per-signal normalisation is mandatory.** CLIP cosines occupy ~0.15–0.35 while attribute/region scores span [0,1]; a raw weighted sum would drown the CLIP signal. Each signal is min-max normalised **across the candidate pool** before weighting; signals that don't apply are dropped and their weight **redistributed**. 2. **Bound attributes must be excluded from `attribute_match`.** That signal is *order-agnostic*: both swap variants reduce to {red, white}, so counting them again pulls swapped queries back **together**, diluting the order-sensitive binding. Excluding them improved measured separation **15.8% → 23.7%**.

4. Measured results

4.1 Compositionality — objective and label-free

For a query **Q** and its **colour-swapped twin Q'**, a bag-of-words retriever cannot tell them apart → `top-k` sets nearly **identical**; a compositional one returns **different** images. So **overlap@k is an inverse proxy for compositional sensitivity, needing no human labels**. Both systems run on the **same index**; only ranking differs. **12 pairs × k=10 = 120 slots**, sized so one image (0.008) cannot dominate the mean.

Pair	CLIP	Ours	Pair	CLIP	Ours
green top / yellow skirt	1.00	0.10 ✓	blue shirt / red trousers	0.90	0.70 ✓
purple top / black trousers	0.70	0.20 ✓	black dress / white jacket	0.90	0.80 ✓
yellow top / blue skirt	0.80	0.30 ✓	black jacket / white trousers	0.70	0.70 =
green jacket / white shirt	0.80	0.40 ✓	pink blouse / black skirt	0.70	0.70 =
brown coat / black trousers	0.90	0.50 ✓	white blouse / blue jeans	0.40	0.50 ✗
red skirt / white blouse	0.70	0.60 ✓	red tie / white shirt	0.50	0.80 ✗
MEAN	0.750	0.525	relative reduction		-30.0%

8/12 wins, 2 ties, 2 losses. Clearest case: *green top / yellow skirt*, where CLIP returns an **identical** top-10 for both phrasings (1.00) — the textbook failure — while this system separates them almost completely (0.10).

Where gains concentrate. All 8 wins are pairs where CLIP is most blind (0.70–1.00); both losses are pairs CLIP *already* separated (0.40, 0.50). **The mechanism contributes most precisely where the baseline fails hardest** — the desired behaviour for a corrective signal, and it suggests making the binding weight *adaptive* to how confidently the dense signals already discriminate.

4.2 Honest negative result, diagnosed

Red tie / white shirt — query 5 — **did not improve**. Traced in data: the index held only **18 tie regions / 3,200 images**, because two filters removed small garments (0.5% area threshold; a 6-region cap applied to an *area-sorted* list, so a tie — the smallest item — was truncated). I lowered the threshold to 0.15% and raised the cap: regions **7,321 → 9,851**, ties **18 → 44** — and query 5 *still* didn't improve.

The remaining ceiling is the corpus, not the algorithm. Fashionpedia is womenswear/runway-heavy: 44 ties (~1.4%), few red. The mechanism demonstrably works where garments exist (1.00 → 0.10 above). The fix is **dataset composition** (blend in menswear/formal imagery), not a ranker change. *Two plausible assumptions* — "regions exist" and "more regions will fix it" — were both false; only measurement revealed it.

4.3 Official queries · corpus

#	Score	Top result	
1	0.942	"a little girl in a yellow raincoat and red tights"	★ exact
2	0.916	"a woman in a black suit and white shirt"	✓ attire right; office scene is the weak axis
3	0.906	"a young girl sitting on a bench in the park"	✓ scene + pose
4	0.896	"on the sidewalk in a white shirt and jeans"	★ inferred from no garment words
5	0.702	"a man in a vest and red pants"	⚠ corpus-limited (§4.2)

Latency **318 ms** mean (~150 ms warm). Zero-shot environment tagging confirms the required axes: urban street **765**, home interior **391**, park **292**, office **198**; remainder catalogue/runway "studio" (1,177) — the Fashionpedia bias, and why query 2's scene component is weakest.

5. Shortcomings and fixes

Shortcoming	Cause	Fix
Small garments under-detected	area threshold + region cap; only 44 ties/3,200	lower threshold (done); accessory-specialised detector; weight by <i>saliency</i> not area
Multi-person images bind wrongly	SegFormer parses clothing, not <i>people</i> — a red tie on A and white shirt on B satisfies both	person detector; require bindings to resolve within one individual
Uncalibrated attribute confidences	softmax gives <i>relative</i> , not probabilistic, scores; scales differ per axis	per-axis Platt/temperature calibration
Scene axis corpus-limited	Fashionpedia is catalogue-heavy (~6% office)	blend a scene-rich people dataset — loader is dataset-agnostic by design
Parser misses negation/comparatives	"no tie", "more formal than" out of grammar	small instruction-tuned LLM emitting the same QuerySpec JSON — interface exists
Hand-set weights	no labelled set in the time budget	learn the fusion (§7.2)

6. Modular code · Scalability · Zero-shot (Criteria 2–4)

Modular — logic separated from data. `configs/*.yaml` holds every knob (no constant in code); `src/models/` wrappers are the **only** code importing HuggingFace, so FashionCLIP → SigLIP is a one-line YAML change; **vectors in FAISS, metadata in SQLite** — **never in filenames**; typed dataclasses (`ImageRecord` / `RegionRecord` / `QuerySpec`) pass between independently testable stages; the loader assumes only "a folder of images", so Fashionpedia/DeepFashion/COCO/phone photos all work unchanged. *Filename-independence is structural: the dataset's filenames are MD5 hashes with zero semantic content, and `image_path` never enters any scoring computation.*

Scalability to 1M+. The retrieval **logic is size-invariant**; only the index type changes. *Recall:* `IndexFlatIP` → `IndexIVFFlat` / HNSW is a **one-line** change in `VectorStore.build(index_type="ivf")` — ids and the `search()` contract are identical, so nothing else moves; $O(N) \rightarrow \sim O(\sqrt{N})/O(\log N)$. *Rerank:* cost is **independent of corpus size** — only the fixed 400-candidate pool is scored — the key scalability property. *Memory:* $1M \times 512\text{-d fp16} \approx \mathbf{1\ GB}$ global (regions $\sim 2.3\times$); IVF-PQ compresses 8–32 $\times$. *Metadata:* SQLite is fine into low millions, and structured attributes enable **SQL pre-filtering** (`environment=office AND EXISTS a tie region`) to shrink the ANN space *before* search. *Indexing:* embarrassingly parallel — shard and merge FAISS shards. Measured **3,200 images in ~11 min** on one 4 GB laptop GPU (~ 0.2 s/image).

Zero-shot. Nothing is trained on this corpus or these queries — FashionCLIP, BLIP, MPNet, SegFormer are all off-the-shelf, and attributes come from **zero-shot CLIP classification** against an editable vocabulary. A new colour, garment, city or weather condition is added by editing `vocab.py` with **no retraining**. Query 4 is the demonstration: "casual weekend outfit for a city walk" has **no garment label**, yet returns sidewalk/jeans imagery because caption and attribute signals infer the vibe rather than matching a keyword.

7. Future work (Deliverable 4)

7.1 Adding locations (cities, places) and weather

Environment is already a **first-class structured axis**, so this is mostly *data* plus new scoring terms — no architectural rewrite.

- Vocabulary extension (no code-path change):** add `weather = [sunny, rainy, snowy, overcast...]` and richer `place` values to `vocab.py` with prompt templates. FashionCLIP tags them **zero-shot** immediately; the parser gains synonyms. This alone makes "a raincoat on a rainy street" prefer genuinely wet scenes.
- Specialised classifiers for precision:** **Places365** for scene/venue and a sky/weather classifier, stored as two more attribute axes and two more weighted terms.

3. **Geo metadata + reverse geocoding:** resolve GPS EXIF to city/landmark. Enables a powerful hybrid — a *hard SQL geo filter* ("in Paris") composed with *soft visual ranking*, exactly the pre-filter pattern from §6.
4. **Landmark recognition** ("in front of the Eiffel Tower") via embedding lookup against a landmark gallery — another region-like signal.
5. **Garment ↔ weather consistency prior:** reward agreement (raincoat ↔ rain, parka ↔ snow), capturing context a query *implies* but doesn't state — enabling "*dressed for bad weather*".

Each is "another axis + another weighted signal", slotting into the existing fusion with no change to the retrieval flow.

7.2 Improving precision

1. **Build a labelled validation set** — highest value. Without labels, tuning is guesswork; with a few hundred (query, relevant_id) pairs we get **precision@k / nDCG** and every change becomes measurable. The harness already accepts `--labels`.
2. **Learn the fusion:** logistic regression or **LambdaMART** over the four signals replaces hand-tuned constants and naturally learns *query-dependent* weighting.
3. **Contrastive fine-tuning on hard negatives:** LoRA-adapt FashionCLIP using **colour-swapped pairs** as explicit negatives, pushing binding into the *encoder* — attacking the root cause in §1.
4. **Stronger colour grounding:** ensemble the CLIP colour head with a classical **CIELAB** dominant-colour estimate per region (robust to lighting) — should lift queries 1, 3, 5.
5. **Per-person binding** (§5) — the biggest correctness fix for crowded scenes.
6. **Cross-encoder reranking** — implemented behind a flag, to be enabled only once its uplift is *measured*.
7. **Query expansion** for style inference: expand "*weekend casual*" into likely garments via a learned style → garment prior.

Appendix — Reproducing

```
pip install -r requirements.txt
python main.py index                # Part A – build FAISS + SQLite
python main.py query "a red tie and a white shirt" -k 10 # Part B – top-k retrieval
python main.py evaluate             # 5 official queries → contact sheet
python main.py compositional        # the colour-swap test of §4.1
python webapp/app.py               # optional visual portal
pytest -q                          # unit tests
```

Stack: FashionCLIP ([patrickjohncyh/fashion-clip](#)) · BLIP ([blip-image-captioning-base](#)) · MPNet ([all-mpnet-base-v2](#)) · SegFormer ([mattdmjaga/segformer_b2_clothes](#)) · FAISS · SQLite. Hardware: one 4 GB NVIDIA RTX 2050.